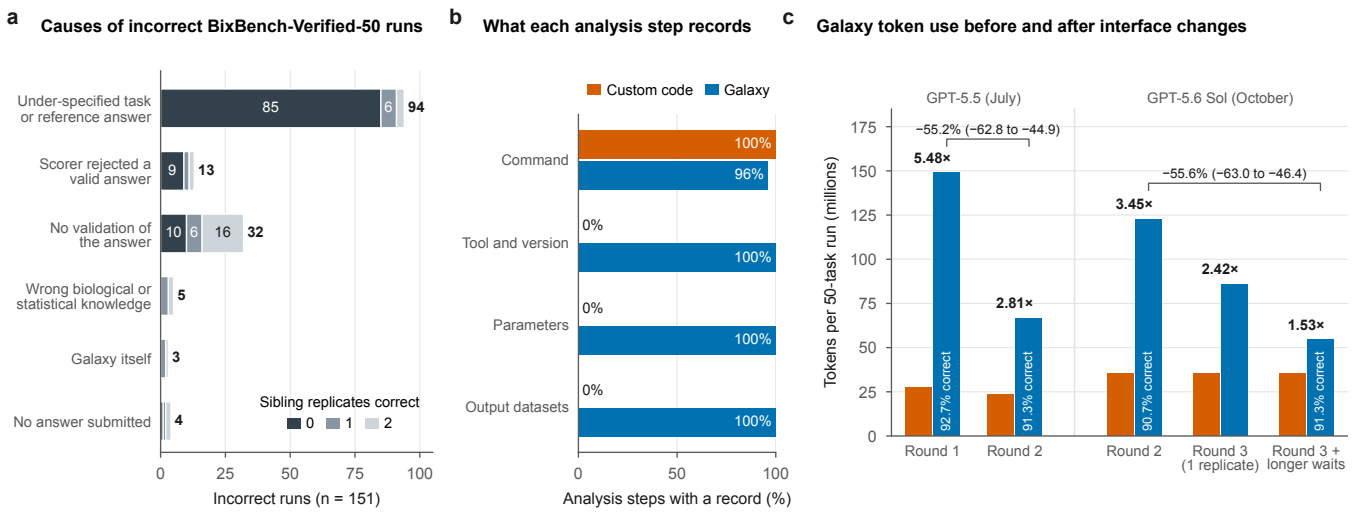

a
Causes of incorrect BixBench-Verified-50 runs
Under-specified task or reference answer
85
6
94
Scorer rejected a valid answer
9
13
No validation of the answer
10
6
16
32
Wrong biological or statistical knowledge
5
Galaxy itself
3
No answer submitted
4
Sibling replicates correct
0
1
2
0
25
50
75
100
Incorrect runs (n = 151)
b
What each analysis step records
Custom code
Galaxy
Command
100%
96%
Tool and version
0%
100%
Parameters
0%
100%
Output datasets
0%
100%
0
50
100
Analysis steps with a record (%)
c
Galaxy token use before and after interface changes
GPT-5.5 (July)
GPT-5.6 Sol (October)
Tokens per 50-task run (millions)
0
25
50
75
100
125
150
175
−55.2% (−62.8 to −44.9)
5.48×
2.81×
−55.6% (−63.0 to −46.4)
3.45×
2.42×
1.53×
92.7% correct
91.3% correct
90.7% correct
91.3% correct
Round 1
Round 2
Round 2
Round 3 (1 replicate)
Round 3 + longer waits